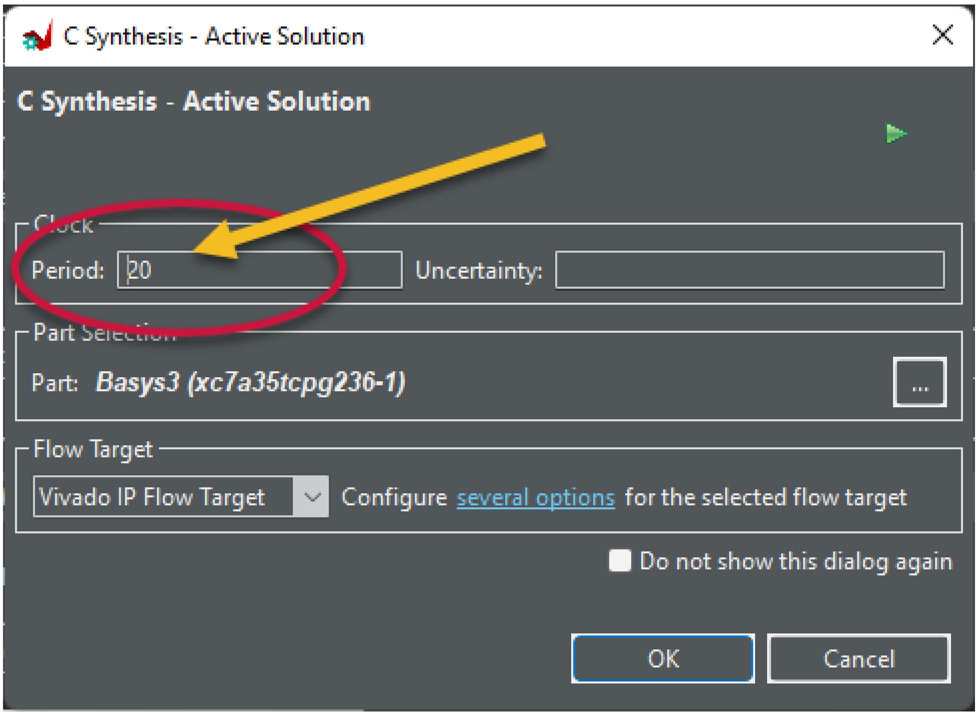
The issue is that it takes two clock cycles to finish. In other words, the design propagation delay is greater than the design clock period.

There are three different solutions for this quiz.

1- The first solution is general and correct for all ranges of a, whether negative or positive.
   As the propagation delay of the circuit is more than the clock period, we can increase the design clock period. For example, replace 10 with 20, as shown in the following figure.



Then design propagation delay would be less than the clock period. Consequently, the final hardware would be a single cycle circuit.

2- The divide-by-4 operator, for a positive number, can be replaced by a 2-bit shift to the right operator that is

a/4 ---------> a >> 2

Note that shift to the right operator can be implemented with a simple hardware circuit with a short propagation delay. Therefore the final hardware would be a single cycle circuit.

```
void expression_quiz(int a, int &b) {
#pragma HLS INTERFACE ap_none port=a
#pragma HLS INTERFACE ap_none port=b
#pragma HLS INTERFACE ap_ctrl_none port=return

    static int s = 0;



    s = s + a >> 2;

    b = s;

}
```

The HLS also uses shift to the right to implement the divide-by-4 operator. However, this replacement cannot be directly applied to negative numbers. As the a input argument's data type is *int*, the HLS tool should provide a correct implementation for both positive and negative numbers. Therefore, it converts the code as follows for synthesis

```
void expression_quiz_shift(int a, int &b) {
#pragma HLS INTERFACE ap_none port=a
#pragma HLS INTERFACE ap_none port=b
#pragma HLS INTERFACE ap_ctrl_none port=return

	static int s = 0;

	int t = a;
	if (a < 0) {
		t = -(-a >> 2);
	}else {
		t = a >> 2;
	}

	s = s + t;

	b = s;
}
```

The extra *if-else* statement in this code increases the design propagation delay, which cannot fit into a single clock period. Therefore, we need two clock cycles to perform the task in the design.

3- In the third solution, as we know a is positive, we can change its data types to *unsigned int*. Then, HLS can simply replace the a/4 with a>>2 expression. Therefore, the propagation delay of the circuit is short enough to be a single cycle design considering the clock period of 10ns.